AB \ CD	00	01	11	10
00			0	
01	0	0	0	0
11	0	0	x	x
10	X			X

$(A+B')\cdot(B'+C')\cdot(A+C'+D')$

$B'\cdot(A+C'+D')$

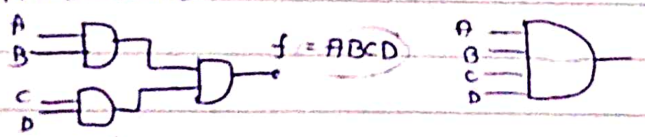
Two level Realization

- consider four gates - AND, OR, NAND, NOR.
- in two level Realization 16 combination are possible.
- this ~~AND~~ TLR are classified into two categories

OR 1) Degenerative form

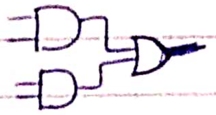
→ if the o/p of two level logic can be obtain by using single logic gate then it is called degenerative form.

Ex AND-AND



2) Non-degenerative

Ex AND-OR



→ List out all degenerative forms -

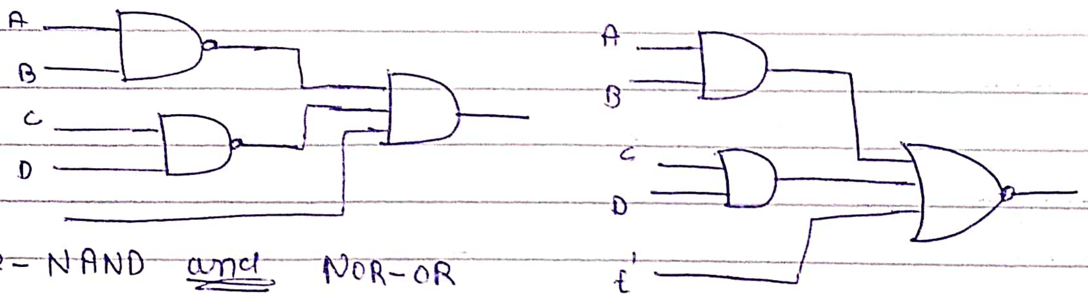
AND-AND, AND-NAND, OR-OR, OR-NOR, NAND-NOR, NOR-NAND

- some non degenerative forms are equivalent to each other.

Ex AND-NOR ~~OR-NAND~~ OR-NAND
and and
NAND-AND ~~NOR-OR~~ NOR-OR

- if complement of some function is given in SOP form than ~~to~~ which can be implemented using above equivalent degenerating form.

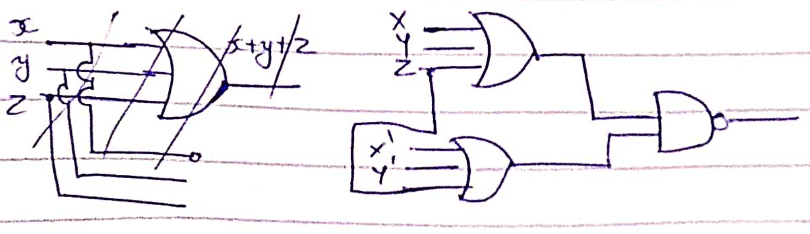
Ex $f' = AB + CD + E'$
$f = (A'+B')(C'+D')E$



OR-NAND and NOR-OR

- if complement of the function is given in POS form then this above ~~dee~~ non-degenerative combination may be use.

Ex $f = x'y'z' + xyz'$ implement using non-degenerative forms.
$f' = (x+y+z)\cdot(x'+y'+z)$
$f = ((x+y+z)(x'+y'+z))'$



Combinational Circuit using MSI & LSI

- In any combinational circuit overall cost is decided by no. of ICs, no. of external internal connection and type of ICs.
- In IC set of components are mounted on a single chip of semiconductor material.
- Based on the no. of components mounted, the integration is classified in 3 types:
  ↳ SSI - Small scale integration, 3-30 gates.
  ↳ MSI - medium " " , 30-300 gates.
  ↳ LSI - Large " " , 3000-3000 gates.
  ↳ VLSI - Very large " " , 3000+ Components.
- Objective of this chapter design a combinational circuit with existing MSI, LSI.

↳ use the combinational circuit to implement boolean function using (1) decoder (2) Multiplexer (3) ROM (4) PLA

4-bit parallel adder.

- ~~Multiple table clustering file organization~~

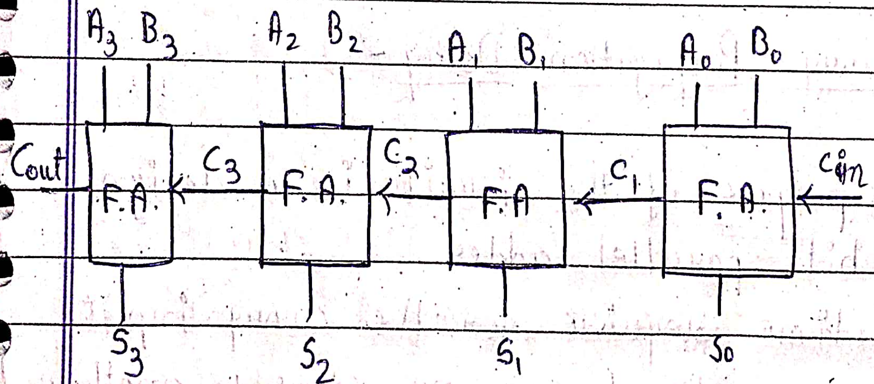
JHB

DDC

ch-5

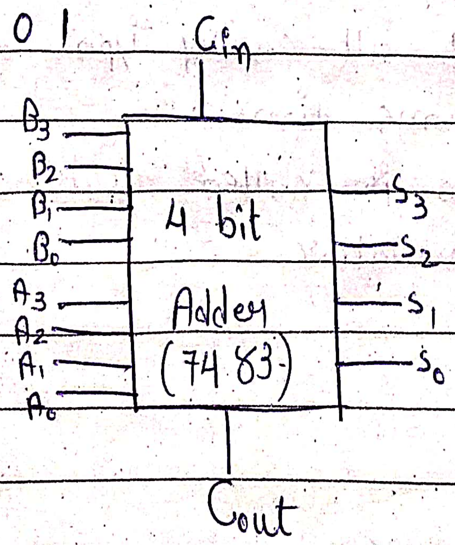
4 bit parallel adder

- to construct 4 bit adder we use 4-full adders, the carry output of each stage is connected to carry input of next stage.

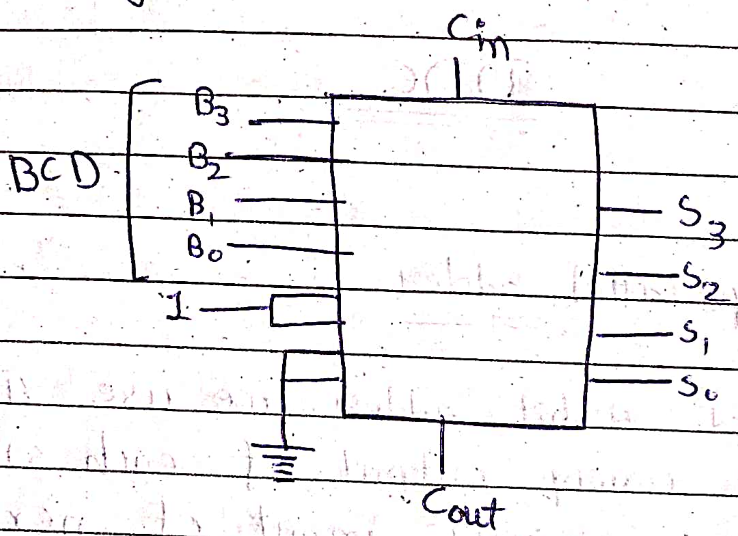


[Ripple carry adder]

$$\begin{array}{r} 1010 \\ 0111 \\ \hline \boxed{1}\,0001 \end{array}$$



Eg:- design a BCD to Excess-3 converter using 4-bit adder MSI.



Carry Propagation Delay -

- the perpose of the circuit is to design 4-bit parallel adder.
- the time require for the carry input to propogate from one stage to another stage it's called carry propogation delay
- We make modification to our circuit such that all carry i/p(s) are calculated at the same time.
- This design is called look a ahead carry generator.

Full adder eq^n

$$C_{out} = \underbrace{AB}_{G_i} + \underbrace{(A \oplus B)}_{P_i} C_{in}$$

$$C_{out} = G_i + P_i C_{in}$$

$$S = \cancel{A \oplus}\ P_i \oplus C_{in}$$

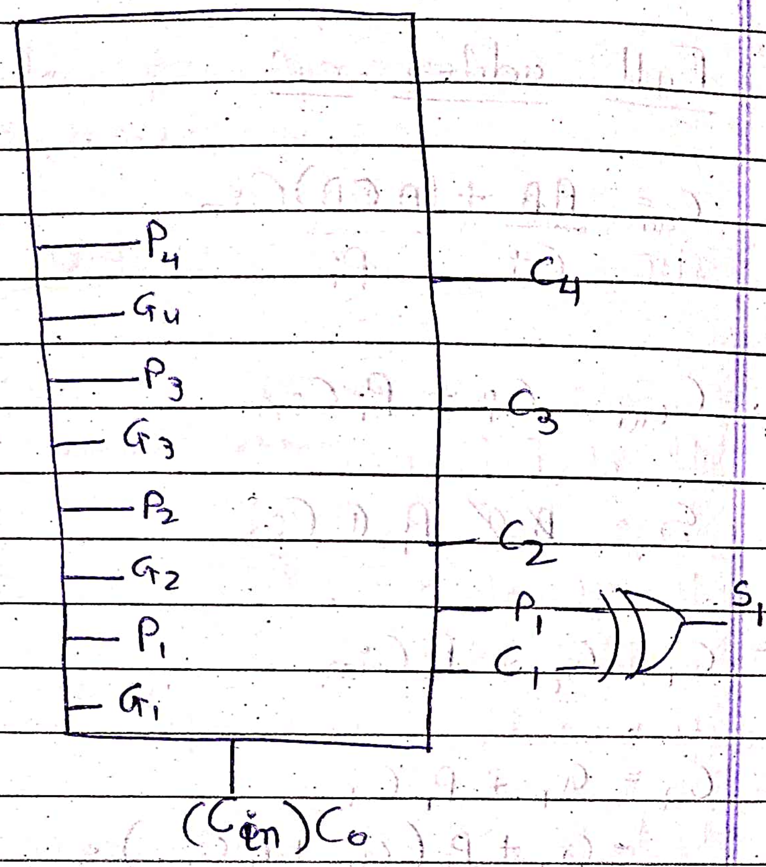
$$C_1 = G_0 + P_0 C_{in}$$

$$\begin{aligned} C_2 &= G_1 + P_1 C_1 \\ &= G_1 + P_1 (G_0 + P_0 C_{in}) \\ &= G_1 + P_1 G_0 + P_1 P_0 C_{in} \end{aligned}$$

$$\begin{aligned} C_3 &= G_2 + P_2 C_2 \\ &= G_2 + P_2 (G_2 + P_1 G_0 + P_1 P_0 C_{in}) \\ &= G_2 + P_2 G_1 + P_1 P_2 G_0 + P_0 P_1 P_2 C_{in} \end{aligned}$$

$$\begin{aligned} C_4 &= G_3 + P_3 C_3 \\ &= G_3 + G_2 P_3 + P_2 P_3 G_1 + P_1 P_2 P_3 G_0 + P_0 P_1 P_2 P_3 C_{in} \end{aligned}$$

MSI Circuit for look ahead carry generator

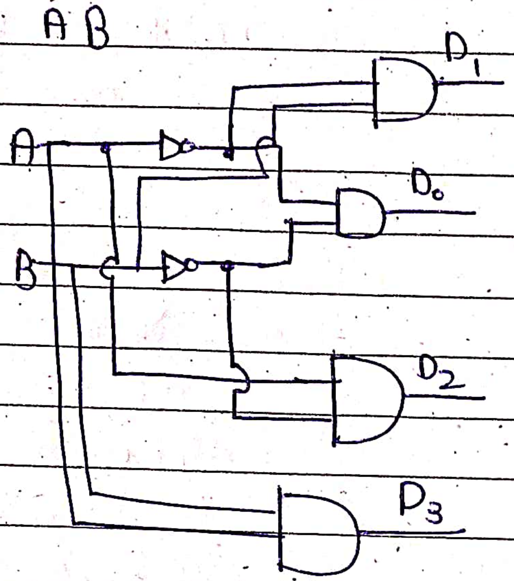


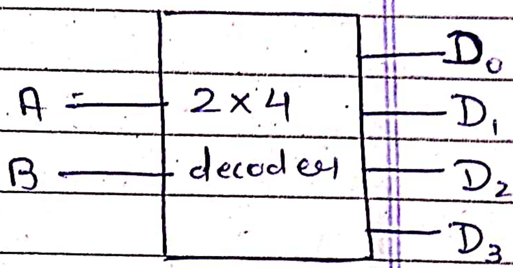
Decoder

- decoder is combinational circuit that converts binary information from n input lines to 2^n output lines.
- dimension of decoder: $n \times 2^n$
- Truth table of 2×4 decoder.

A	B	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

$D_0 = A'B'$
$D_1 = A'B$
$D_2 = AB'$
$D_3 = AB$





(3 x 8) Truth table

A	B	C	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1